

Program : Diploma in Civil and Environmental Engineering	
Course Code : 4356	Course Title: Environmental Chemistry and Microbiology Lab
Semester : 4	Credits: 1.5
Course Category: Programme Core Course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To apply principles of chemistry and microbiology to determine quality parameters of water and waste water

Course Prerequisites:

Topic	Course code	Course name	Semester
	1004	Applied Chemistry	1
	2001	Environmental Science	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Apply volumetric analysis to determine quality parameters of water.	12	Applying
CO2	Apply principles of gravimetric analysis, colorimetry, and nephelometry to determine quality parameters of water.	6	Applying
CO3	Determine optimum coagulant dosage, and chlorine demand of given water sample.	6	Applying
CO4	Apply principles of chemistry and microbiology to determine oxygen demand and MPN of waste water and surface water.	12	Applying
	Open ended experiments	3	Applying
	Lab tests	6	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3	2		
CO2	3			3	2		
CO3	3			3	2		
CO4	3			3	2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Apply volumetric analysis to determine quality parameters of water.		
M1.01	Determination of acidity of given water sample.	3	Applying
M1.02	Determination of alkalinity of given water sample.	3	Applying
M1.03	Determination chloride content of given water sample.	3	Applying
M1.04	Determination of DO of given water sample	3	Applying
CO2	Apply principles of gravimetric analysis, colorimetry, and nephelometry to determine quality parameters of water.		
M2.01	Determination of turbidity of given water sample.	1	Applying
M2.04	Determination of total solids, dissolved solids, suspended solids and volatile solids in water.	2	Applying
M2.03	Determination of iron content of given water sample using spectrophotometer.	3	Applying
	Lab test 1	3	
CO3	Determine optimum coagulant dosage, and chlorine demand of given water sample.		
M3.01	Determine Optimum coagulation dosage by conducting jar test.	3	Applying

M3.02	Determine chlorine demand of given water sample.	3	Applying
CO4	Apply principles of chemistry and microbiology to determine oxygen demand and MPN of waste water and surface water.		
M4.01	Determine BOD of given sample of sewage or surface water.	3	Applying
M4.02	Determine COD of given sample of sewage or surface water.	3	Applying
M4.03	Determine total coliforms in the given sample of surface water or underground water using multiple tube fermentation technique.	6	Applying
	Open ended experiments	3	Applying
	Lab test- II	3	

Text / Reference:

T/R	Book Title/Author
R	Clair N Sawyer, Perry L McCarty, Gene F Parkin- Chemistry for Environmental Engineering and Science- Mc Graw Hill
R	Ian L Pepper, Charles P Gerba, Terry J Gentry, Environmental Microbiology, Academic Press
R	F.W.Fifield and P J Haines -Environmental Analytical Chemistry- Blackwell Science
R	IS 3025- Methods of sampling and tests of water and waste water.
R	Standard methods for the examination of water and wastewater-American Public Health Association; American Water Works Association; and Water Pollution Control Federation U.S.A
R	IS 1622-Methods of sampling and microbiological examination or water

Online resources:

Sl.No.	Website Link
1	NITTTR NCTEL (nitttrchd.ac.in)
2	Welcome to Virtual Labs - A MHRD Govt of india Initiative (vlabs.ac.in)
3	Welcome to Virtual Labs - A MHRD Govt of india Initiative (vlabs.ac.in)